

Proof of Arden's Lemma and Extended Arden's Lemma

Tutorial 2 — Automata Theory

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1 Arden's Lemma

Lemma 1 (Arden). *If $R = Q + RP$ with $\varepsilon \notin P$, then $R = QP^*$.*

Proof. We first show that $QP^* \subseteq R$. To do this, we show that $QP^j \subseteq R$ for all j ; the required result then follows immediately.

We proceed by induction. For $j = 0$, we have $Q \subseteq Q + RP = R$, so the base case holds. Assume it holds for j . Then for $j + 1$,

$$QP^{j+1} = QP^jP \subseteq RP \subseteq Q + RP = R,$$

which finishes the induction step.

Now for the converse, we want to show $R \subseteq QP^*$. We induct on the length of words of R . Let $x \in R$.

For the base case, we check all words of the smallest length in R . Let this length be a . Then $x \in R$ implies $x \in Q$ or $x \in RP$. If $x \in Q$, then clearly $x \in QP^*$. If $x \in RP$, then $x = yz$ with $y \in R$, $z \in P$. But since $\varepsilon \notin P$ we have $z \neq \varepsilon$, and thus $|y| < |x| = a$, which is a contradiction. Thus this case cannot occur. This finishes the base case.

Assume we have shown this for all words of length at most n . Let $x \in R$ with $|x| = n + 1$. Again $x \in Q$ or $x \in RP$. If $x \in Q$, we are done. If $x \in RP$, then $x = yz$ with $y \in R$, $z \in P$, again with $z \neq \varepsilon$. Thus $|y| < |x| = n + 1$, implying $|y| \leq n$ and hence $y \in QP^*$. Thus $x = yz \in QP^*P \subseteq QP^*$, which finishes the proof of the converse.

By basic set theory, $R = QP^*$. □

2 Extended Arden's Lemma

Remark. If $\varepsilon \in P$, then there are more solutions, but they can be characterized easily.

Theorem 1 (Extended Arden). *If $R = Q + RP$ and $\varepsilon \in P$, then the solution set of R is*

$$\{ AP^* \mid A \supseteq Q \}.$$

Proof. This is actually simpler. First we show that any element of that set is a solution. Since $\varepsilon \in P$, we have $P^* = P^+$, implying $AP^*P = AP^+ = AP^*$. Also $Q \subseteq A \subseteq AP^*$, and thus

$$Q + RP = Q + AP^*P = Q + AP^* = AP^* = R.$$

Conversely, we show that these are all solutions. Suppose B is a solution. Then $B = Q + BP$, implying $Q \subseteq B$. Also, since $\varepsilon \in P$, we have $Q \subseteq B \subseteq BP$, and hence $B = Q + BP = BP$. By induction $B = BP^n$ for all n , implying $B = BP^*$. Since $B \supseteq Q$, this lies inside that set. We are done. $\square$